Point vs interval estimation + estimator characteristics

C8.01-C8.68 CORE ONLY (SKILL IDs)

1 RECOGNISE IT

- Asked to distinguish one estimate from a range.
- Need to judge an estimator by repeated-sampling behaviour.
- Parameter is unknown; statistic estimates it.

2 DECISION / METHOD

1. Match parameter to statistic: μ <- X̄; π <- p.
2. Point estimate = one statistic value.
3. Interval estimate = plausible range plus confidence level.
4. Judge unbiasedness, consistency and relative efficiency.

3 FORMULA

```
unbiased: E(estimator)=parameter
consistent: error shrinks as n grows
more efficient: smaller variance
among unbiased estimators
```

4 FORMULA SHEET VS WHAT I KNOW

Gives: parameter/statistic pairings and standard errors.
Still know: definitions of point, interval, unbiasedness, consistency and efficiency.

5 EXCEL ROUTE

- Mean point estimate: `=AVERAGE(range)`
- n: `=COUNT(range)`
- Proportion point estimate: successes/count.

6 TINY WORKED EXAMPLE

A sample gives X̄=42.6. That is a point estimate of μ. A 95% CI such as (40.1,45.1) adds precision information.

7 TOP EXAM TRAPS

- Saying the parameter changes from sample to sample.
- Unbiased means 'always exactly correct'.
- Calling a confidence level a probability that this fixed interval is correct.

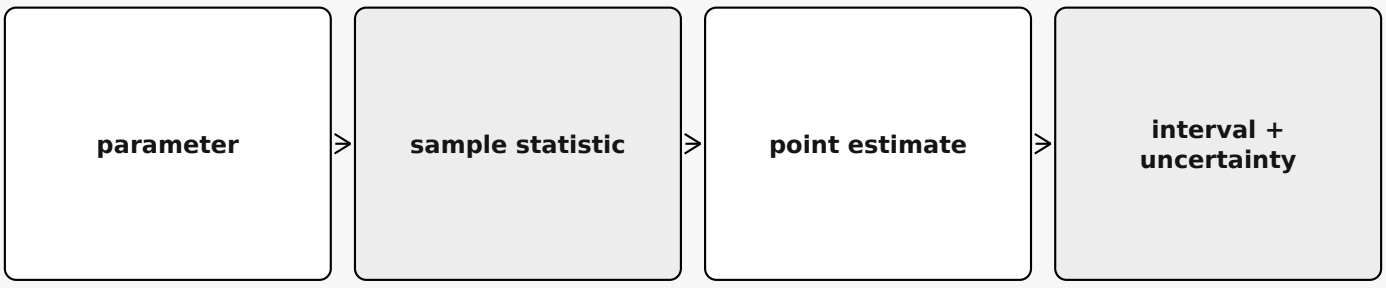
8 SAY ALOUD

The statistic is my sample-based point estimate; the confidence interval shows its uncertainty about the fixed parameter.

9 EXIT CHECK

Which estimator characteristic prefers smaller variance when both estimators are unbiased?

ONE-GLANCE VISUAL ROUTE



Confidence, α, critical value, SE and margin of error

C8.01-C8.68 CORE ONLY (SKILL IDs)

1 RECOGNISE IT

- Any confidence-interval question before selecting a parameter-specific method.
- Confidence level C is supplied.
- Need critical value, SE, margin or endpoints.

2 DECISION / METHOD

1. α=1-C.
2. For a two-sided CI, place α/2 in each tail.
3. Select Z or t.
4. Compute SE.
5. Margin e*=critical×SE; CI=estimate±e*.

3 FORMULA

$$CI = \text{estimate} \pm \text{critical} \times SE$$
$$\alpha = 1 - C$$
$$z^* = \Phi^{-1}(1-\alpha/2) > 0$$
$$t^* = T.INV.2T(\alpha, df) > 0$$
$$\text{margin} = \text{critical} \times SE$$

4 FORMULA SHEET VS WHAT I KNOW

Gives: parameter-specific SE formulas.
Still know: general CI structure, confidence-to-α conversion, and when critical is Z versus t.

5 EXCEL ROUTE

- Z critical: `=NORM.S.INV(1-alpha/2)`
- t critical: `=T.INV.2T(alpha,df)`
- endpoints: estimate ± critical*SE.

6 TINY WORKED EXAMPLE

For C=.90, α=.10 and each tail=.05. The positive Z critical is NORM.S.INV(.95)=1.6449.

7 TOP EXAM TRAPS

- Using α instead of α/2 for a two-sided CI.
- Confusing confidence level with critical value.
- Using SD where SE is required.

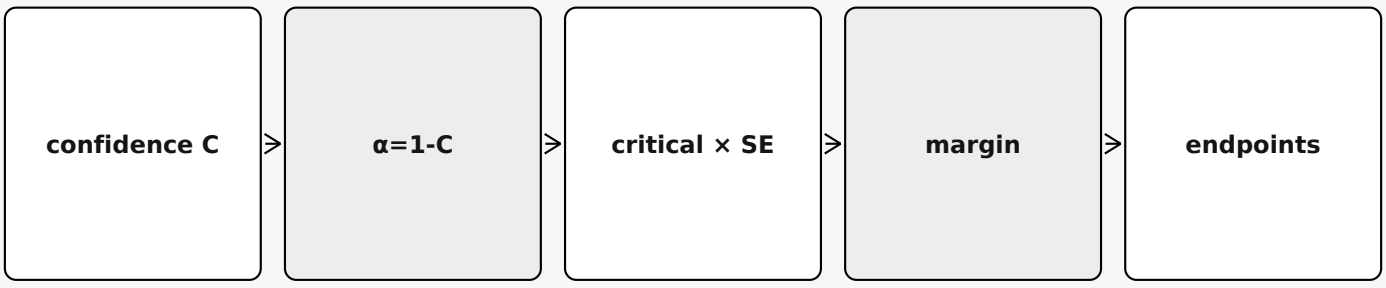
8 SAY ALOUD

Every confidence interval is point estimate plus or minus a critical value times a standard error.

9 EXIT CHECK

For 98% confidence, what cumulative-left area gives the positive Z critical?

ONE-GLANCE VISUAL ROUTE



Mean CI, σ known -> Z

C8.01-C8.68 CORE ONLY (SKILL IDs)

1 RECOGNISE IT

- Estimate population mean μ.
- Population σ is explicitly supplied/known.
- Random sample; population normal or sample large enough.

2 DECISION / METHOD

1. Compute $\bar{X}$ and n.
2. SE=σ/√n.
3. z*=NORM.S.INV(1-α/2) > 0.
4. $\bar{X}$±z*SE.
5. Report units and confidence.

3 FORMULA

$z^* = \Phi^{-1}(1-\alpha/2) > 0$

$CI: \bar{X} \pm z^* \times \sigma/\sqrt{n}$

4 FORMULA SHEET VS WHAT I KNOW

Gives: $\bar{X}$, SE=σ/√n, Z lane when σ known. Still know: critical value, CI endpoints and interpretation.

5 EXCEL ROUTE

- `n=COUNT(range); Xbar=AVERAGE(range)`
- `zstar=NORM.S.INV(1-alpha/2)`
- lower/upper = `Xbar +/- zstar*sigma/SQRT(n)`.

6 TINY WORKED EXAMPLE

$\bar{X}$=40,σ=6,n=36,C=.95. SE=1 and z=1.96, so CI is `(38.04,41.96)`.

7 TOP EXAM TRAPS

- Using sample s when σ is supplied.
- Using t in the known-σ lane.
- Treating the CI as a range for individual X values.

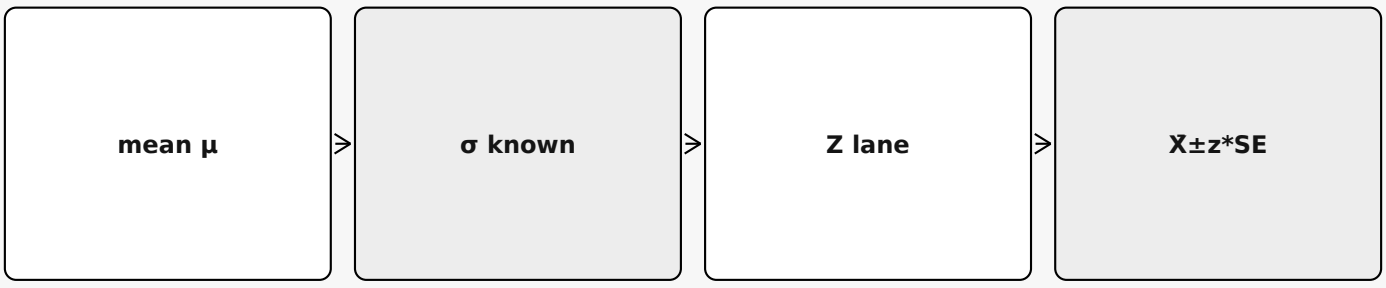
8 SAY ALOUD

Known population σ means a Z interval for μ centred at $\bar{X}$ with SE σ/√n.

9 EXIT CHECK

What changes in the interval if n quadruples but everything else stays fixed?

ONE-GLANCE VISUAL ROUTE



Student t, df and mean CI when σ unknown

C8.01-C8.68 CORE ONLY (SKILL IDs)

1 RECOGNISE IT

- Estimate population mean μ.
- Population σ unknown; sample s available.
- Population approximately normal under the course conditions.

2 DECISION / METHOD

1. Compute n, $\bar{X}$, s.
2. df=n-1.
3. t*=T.INV.2T(α,df).
4. SE=s/√n.
5. $\bar{X}$ ± t*SE; interpret subject to normality.

3 FORMULA

```
t* = T.INV.2T(α,df) > 0
CI: X¯ ± t* × s/√n
df=n-1
```

4 FORMULA SHEET VS WHAT I KNOW

Gives: s/√n and t_(n-1) when σ unknown. Still know: t critical lookup/function, normality condition and endpoint construction.

5 EXCEL ROUTE

- `n=COUNT(range)`; `Xbar=AVERAGE(range)`; `s=STDEV.S(range)`
- `tstar=T.INV.2T(alpha,n-1)`
- endpoints: `Xbar +/- tstar*s/SQRT(n)`.

6 TINY WORKED EXAMPLE

$\bar{X}$=25,s=4,n=16,C=.95. df=15,t*=2.131,SE=1 -> CI `(22.869,27.131)`.

7 TOP EXAM TRAPS

- df=n.
- Using Z because n seems moderate.
- Ignoring normality for a small sample.
- Mixing σ and s.

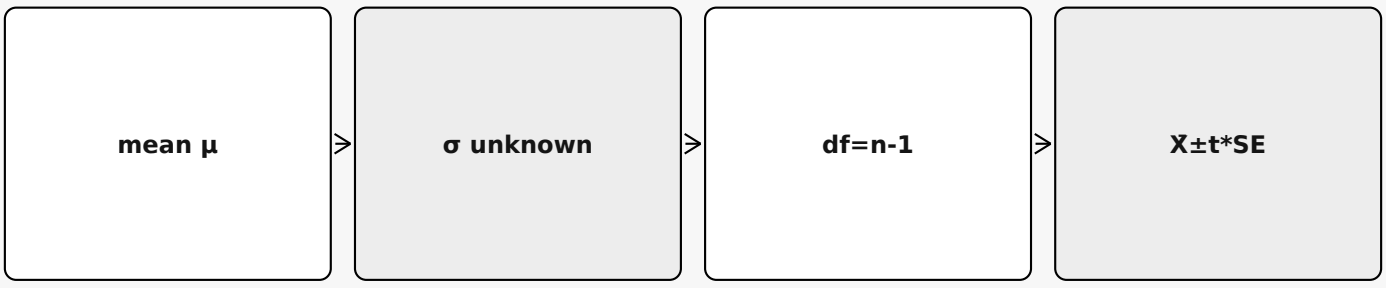
8 SAY ALOUD

Unknown σ means Student t with sample s and n-1 degrees of freedom.

9 EXIT CHECK

Which Excel function returns the positive two-sided t critical value?

ONE-GLANCE VISUAL ROUTE



Population-proportion confidence interval

C8.01-C8.68 CORE ONLY (SKILL IDs)

1 RECOGNISE IT

- Estimate a population percentage/proportion π.
- Sample has x successes out of n.
- Need a two-sided CI and normal approximation is adequate.

2 DECISION / METHOD

1. p=x/n.
2. Check np > 5 AND n(1-p) > 5.
3. Estimated SE=√[p(1-p)/n].
4. CI=p±z*SE.
5. Convert to percent only if requested.

3 FORMULA

p=x/n
SE_hat=√[p(1-p)/n]
$z^* = \Phi^{-1}(1-\alpha/2) > 0$
CI: p ± z*SE_hat

4 FORMULA SHEET VS WHAT I KNOW

Gives: theoretical SE uses π for a sampling distribution.
Still know: a CI estimates unknown π, so substitute sample p into estimated SE.

5 EXCEL ROUTE

- p=successes/n
- SE=SQRT(p*(1-p)/n)
- z=NORM.S.INV(1-alpha/2)
- endpoints p ± z*SE.

6 TINY WORKED EXAMPLE

x=60,n=100,C=.95: p=.60,SE=.0490,margin=.0960 -> CI (.5040,.6960).

7 TOP EXAM TRAPS

- Weakening the strict sourced course inequality.
- Using null π_0 in a CI SE.
- Forgetting p is a proportion, not a percentage.
- Using π instead of sample p in the estimated CI SE.

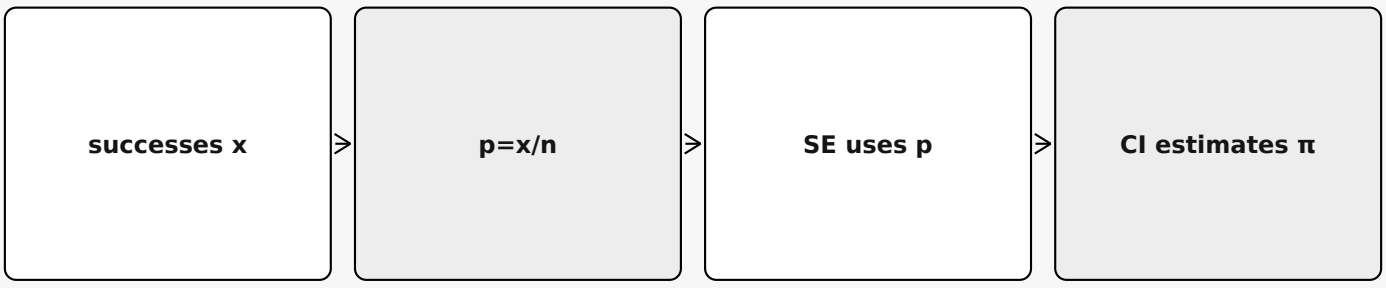
8 SAY ALOUD

A proportion CI uses sample p both as the centre and inside the estimated standard error.

9 EXIT CHECK

Why does a CI for π use p rather than an unknown π inside its estimated SE?

ONE-GLANCE VISUAL ROUTE



Required sample size for a mean

C8.01-C8.68 CORE ONLY (SKILL IDs)

1 RECOGNISE IT

- Need n to estimate μ within margin e at confidence C.
- Population σ, pilot s, or conservative expected SD is supplied.
- This is planning for precision, not a completed CI.

2 DECISION / METHOD

1. Convert C to z*.
2. Choose σ estimate.
3. Compute $n0=(z^*\sigma/e)^2$.
4. ALWAYS ROUND UP.
5. Apply finite-population adjustment only if separately triggered.

3 FORMULA

$z^* = \Phi^{-1}(1-\alpha/2) > 0$

$n_0 = (z^*\sigma/e)^2$

ALWAYS ROUND UP

4 FORMULA SHEET VS WHAT I KNOW

Gives: required-mean sample-size formula.
Still know: how to get z, estimate σ if unknown, and always round up.

5 EXCEL ROUTE

```
=ROUNDUP((NORM.S.INV(1-alpha/2)^2*sigma^2)/e^2,0)
```

6 TINY WORKED EXAMPLE

C=.95,σ=10,e=3:
$n0=(1.96*10/3)^2=42.684$, so required n=43.

7 TOP EXAM TRAPS

- Rounding to nearest/down.
- Using full interval width instead of margin e.
- Using t critical in this planning formula.
- Forgetting FPC is a later step.

8 SAY ALOUD

Mean sample size comes from the margin equation and is always rounded up.

9 EXIT CHECK

If the allowed margin halves, by what factor does required n change?

ONE-GLANCE VISUAL ROUTE



Required sample size for a proportion

C8.01-C8.68 CORE ONLY (SKILL IDs)

1 RECOGNISE IT

- Need n to estimate π within margin e at confidence C.
- Prior/pilot proportion is supplied, or none is available.
- Planning for a future proportion estimate.

2 DECISION / METHOD

1. Get z* from confidence.
2. Choose π estimate: pilot/prior; if none, use .50.
3. Compute n0.
4. ALWAYS ROUND UP.
5. Adjust for finite N only if required.

3 FORMULA

$z^* = \Phi^{-1}(1-\alpha/2) > 0$

$n_0 = z^{*2}\pi(1-\pi)/e^2$

no estimate → π=.50

ALWAYS ROUND UP

4 FORMULA SHEET VS WHAT I KNOW

Gives: required-proportion sample-size formula.

Still know: .50 is conservative, convert percent margin to decimal, and round up.

5 EXCEL ROUTE

```
=ROUNDUP((NORM.S.INV(1-alpha/2)^2*pi*(1-pi))/e^2,0)
```

6 TINY WORKED EXAMPLE

C=.95, prior π=.40,e=.05: n0=368.79, so n=369.

7 TOP EXAM TRAPS

- Using observed x/n when no pilot exists instead of .50.
- Entering e=5 instead of .05.
- Rounding down.
- Confusing CI SE p with planning π.

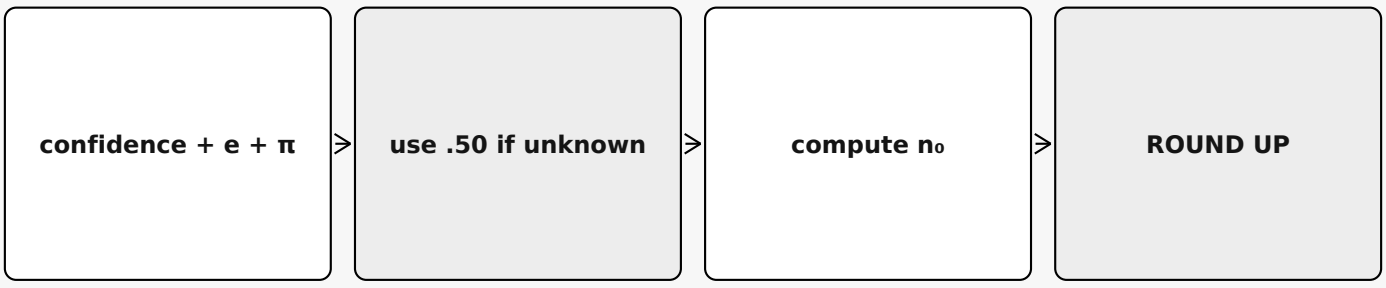
8 SAY ALOUD

For proportion sample size, use the best prior π or conservative .50, then always round up.

9 EXIT CHECK

Why does π=.50 produce the largest planned n?

ONE-GLANCE VISUAL ROUTE



Finite-population correction for confidence intervals

C8.01-C8.68 CORE ONLY (SKILL IDs)

1 RECOGNISE IT

- Finite N, sample without replacement, and n/N>0.05.
- Need a CI for μ or π.
- Sampling a material fraction reduces uncertainty.

2 DECISION / METHOD

1. Build the ordinary SE.
2. FPC=sqrt((N-n)/(N-1)).
3. adjusted SE=ordinary SE*FPC.
4. Use the usual Z/t critical and centre.
5. No FPC for replacement or negligible fraction.

3 FORMULA

FPC=√[(N-n)/(N-1)]
mean CI: $\bar{X} \pm \text{critical} \times (s/\sqrt{n}) \times FPC$
proportion CI: $p \pm z^*\sqrt{[p(1-p)/n]} \times FPC$

4 FORMULA SHEET VS WHAT I KNOW

Gives: finite-population sample-size adjustment only.
Still know: CI FPC multiplier, trigger, and replacement caveat.

5 EXCEL ROUTE

- FPC=SQRT((N-n)/(N-1))
- multiply ordinary SE by FPC before the margin.
- retain Z/t choice from the underlying CI method.

6 TINY WORKED EXAMPLE

N=1000,n=100,p=.30,C=.90: FPC=.9492; adjusted SE=.0435. Using full precision, margin=.0715 and CI≈(.2285,.3715).

7 TOP EXAM TRAPS

- Applying FPC to the centre.
- Dividing by FPC.
- Applying it when sampling with replacement.
- Forgetting the n/N trigger.

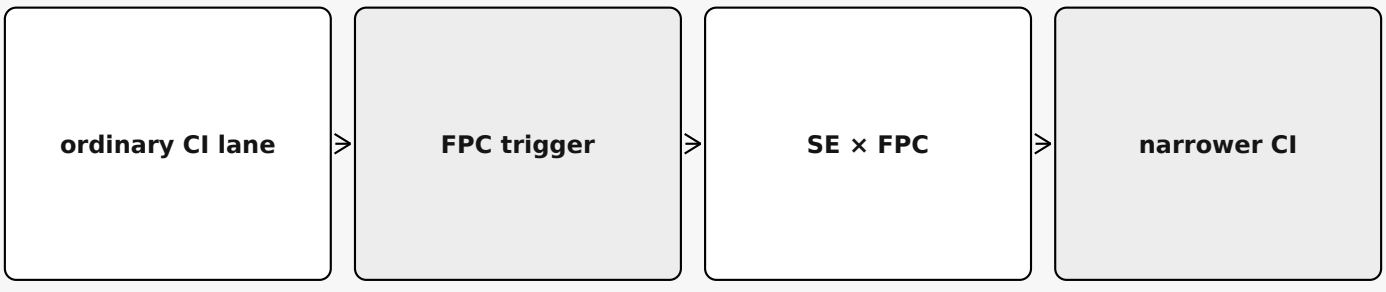
8 SAY ALOUD

For material sampling without replacement, FPC multiplies the ordinary standard error and narrows the interval.

9 EXIT CHECK

Where in the CI formula does FPC belong?

ONE-GLANCE VISUAL ROUTE



Finite-population sample-size adjustment

C8.01-C8.68 CORE ONLY (SKILL IDs)

1 RECOGNISE IT

- Population size N is finite.
- Initial large-population requirement n0 has been calculated.
- Need the final sample size for sampling without replacement.

2 DECISION / METHOD

1. Compute n0 using the ordinary mean/proportion formula.
2. Apply the finite-N adjustment.
3. ALWAYS ROUND UP the adjusted result.
4. Keep n0 and final n distinct.

3 FORMULA

```
n = n0*N / (n0 + (N-1))
```

4 FORMULA SHEET VS WHAT I KNOW

Gives: this finite-population sample-size adjustment.
Still know: first calculate n0 correctly and round the adjusted n up.

5 EXCEL ROUTE

```
=ROUNDUP((n0*N)/(n0+(N-1)),0)
```

6 TINY WORKED EXAMPLE

If n0=385 and N=2000, adjusted n=322.99, so required n=323.

7 TOP EXAM TRAPS

- Applying FPC before n0 exists.
- Using current sample n where formula needs n0.
- Rounding down.
- Applying both SE-FPC and n-adjustment to the same planning step.

8 SAY ALOUD

First find the large-population n0, then shrink it for finite N and round up.

9 EXIT CHECK

Which value enters the adjustment numerator: current n or unadjusted n0?

ONE-GLANCE VISUAL ROUTE



Bootstrapping

C8.01-C8.68 CORE ONLY (SKILL IDs)

1 RECOGNISE IT

- Small sample and population distribution unknown/non-normal.
- Need a CI for a statistic without relying on the standard normal/t route.
- A resampling table or index is supplied.

2 DECISION / METHOD

1. Start with the observed sample.
2. Resample n values FROM THAT SAMPLE WITH replacement.
3. Compute the statistic for every resample.
4. Repeat many times and order the statistics.
5. Use the cutoff convention stated/supported by the course or question.

3 FORMULA

`bootstrap CI = lower and upper empirical quantiles of resampled statistics`

4 FORMULA SHEET VS WHAT I KNOW

Gives: no bootstrap formula.
Still know: resampling source, replacement, same n, many repetitions, statistic and tail cutoffs.

5 EXCEL ROUTE

- Go to the named Bootstrap worksheet and identify the original sample/index grid.
- One mean per resample: =AVERAGE(`resample_range`).
- Order/read the supplied resample means; use stated cutoffs.
- Preserve precision; enter requested limits to four decimals.

6 TINY WORKED EXAMPLE

Observed sample {2,4,7}. One valid n=3 resample is {4,4,2}; its bootstrap mean is 3.333. Repeat to form the bootstrap distribution.

7 TOP EXAM TRAPS

- Resampling from the population/raw database rather than the observed sample.
- Sampling without replacement or changing n.
- Treating one resample as the CI.
- Inventing a universal 250-resample cutoff-index rule.

8 SAY ALOUD

Bootstrap by repeatedly sampling with replacement from the observed sample, then use the empirical statistic distribution.

9 EXIT CHECK

Where do bootstrap draws come from, and are they drawn with replacement?

2026 A1S2 WORKBOOK WORKFLOW

1. Named Bootstrap sheet	2. Find original sample	3. Read supplied index/grid
4. Resample WITH replacement	5. One mean per resample	6. Order resample means
7. Use stated cutoff convention	8. No universal 250 rule	9. Enter 4 decimals

CI interpretation/reporting + method-selection tree

C8.01-C8.68 CORE ONLY (SKILL IDs)

1 RECOGNISE IT

- Need to choose among Z, t, proportion, FPC or bootstrap.
- Asked to interpret/report a finished interval.
- Closest competitor depends on parameter, σ status, normality and finite sampling.

2 DECISION / METHOD

Tree:
1. Parameter: mean or proportion?
2. Mean: σ known → Z; σ unknown → t.
3. Proportion → Z CI using p.
4. If approximate normality is doubtful / the population is clearly non-normal, use bootstrap where the course/question supports it; small n makes the normality issue especially important.
5. Finite without replacement and n/N>.05 → multiply SE by FPC.

3 FORMULA

```
report: estimate, lower, upper,
confidence level, n, units/context
CI describes parameter uncertainty,
not individual observations
```

4 FORMULA SHEET VS WHAT I KNOW

Gives: method-specific SEs and sample-size formulas.
Still know: selection tree and correct repeated-sampling interpretation.

5 EXCEL ROUTE

Use the chosen route:
- Z: NORM.S.INV
- t: T.INV.2T, STDEV.S, df=n-1
- proportion: SQRT(p*(1-p)/n)
- FPC: SQRT((N-n)/(N-1))
- bootstrap: resample and empirical cutoffs.

6 TINY WORKED EXAMPLE

A 95% CI for μ is (12.4,15.1). Say: 'We are 95% confident the population mean lies between 12.4 and 15.1 units.'

7 TOP EXAM TRAPS

- '95% of data lie in the CI.'
- 'There is a 95% probability fixed μ is in this computed interval.'
- Using t for a proportion.
- Forgetting FPC/normality checks.

8 SAY ALOUD

Choose by parameter and σ status, then interpret the interval as uncertainty about the population parameter.

9 EXIT CHECK

Unknown σ, small normal sample, mean parameter: which distribution and df?

ONE-GLANCE VISUAL ROUTE

